

1. A. Available = Total – Allocated = 8 – 6 = 2 Only P1 and P2 can be allocated first.

The safe sequences from the state shown above are:

P2, P1, P3, P4

P2, P1, P4, P3

P2, P3, P1, P4

P2, P3, P4, P1

P1, P2, P3, P4

P1, P2, P4, P3

P1, P3, P2, P4

P1, P3, P4, P2

- B. Yes – Since the request can be granted. If P4 is given 1 resource, the required sequence will be 2,1,3,3 respectively for P1, P2, P3, P4.

Available = Total – Allocated = 8 – 7 = 1

Only P2 can be allocated first.

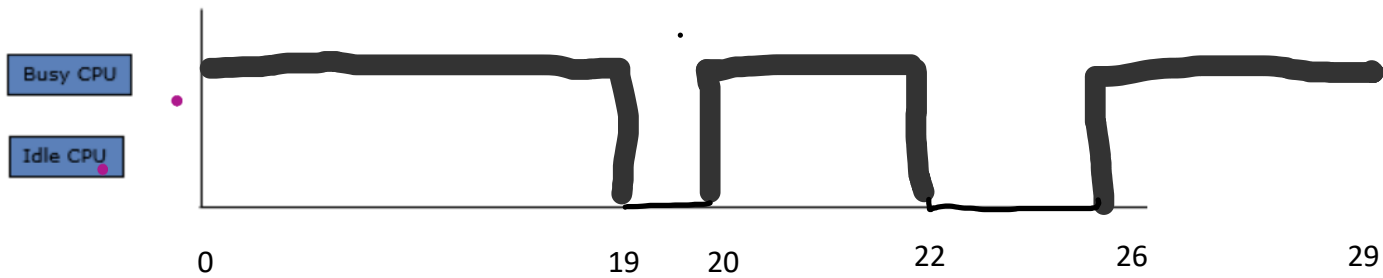
P2, P1, P3, P4 and P2, P1, P3, P4

2. A.

The CPU is assigned in a round robin fashion, with each process getting the CPU for 3 time units before the next process is given the CPU.

Round Robin scheduling with time slice of 3 units:

- 0-3 time units: CPU to P1 for it's first time slice
- 3-6 time units: CPU to P2 for it's first time slice
- 6-9 time units: CPU to P3 for it's first time slice
- 9-12 time units: CPU re-assigned to P1 for its second time slice but it'll only run for 2 seconds before it gets interrupted by P2 because its finished its I/O
- 11-13 time units: CPU completes its I/O and will get assigned to P@ for its second time slice and completes its processing.
- 13-16 time units: CPU assigned to P3 for its second time slice and then will go for its I/O.
- 16-19 time units: CPU assigned to P1 and will run for 3 time units and it'll then go for its I/O
- 19-20 time units: CPU will go into idle sate since both P1 and P3 are running their I/O
- 20-22 time units: CPU assigned to P3 which will run for 2 seconds and then halt.
- 22-26 time units: CPU will idle for 4 seconds because P1 is still running its I/O and P2 and P3 are halted.
- 26-29 time units: CPU will be assigned to P1 and will run for 3 time units and then P1 will be halted as well.



B. Here is the timeline showing all the states P1 goes through until it halts:

- 0-3 time units: P1 is in the Running state on the CPU
- 3-9 time units: P1 is in the Ready state, waiting for the CPU for 6 time units
- 9-11 time units: P1 is back in the Running state on the CPU for 2 time units
- 11-16 time units: P1 is again in the Ready state, waiting for the CPU for 5 time units
- 16-19 time units: P1 is Running on the CPU for 3 time units
- 19-26 time units: P1 is in the Blocked/Waiting state, performing I/O for 7 time units
- 26-29 time units: P1 is Running on the CPU for 3 time units
- After 29 time units: P1 enters the Halted state and completes its execution.
- The timeline for P1 is:
Running (3 units) -> Ready (6 units) -> Running (2 units) -> Ready (5 units) ->
Running (3 units) -> Blocked (7 units) -> Running (3 units) -> Halted

